

E-Commerce Business Intelligence Analysis using PostgreSQL

Project Overview

This project analyzes an e-commerce transactional dataset using PostgreSQL to uncover insights related to revenue performance, customer behavior, product demand, and delivery operations. The objective of this analysis is to demonstrate how SQL and relational databases can transform raw transactional data into meaningful business intelligence insights that support data-driven decision making.

Objectives

- Design a relational database schema for an e-commerce dataset
- Load large transactional datasets into PostgreSQL
- Perform SQL-based exploratory data analysis
- Calculate important business KPIs
- Extract insights related to customer behavior, revenue drivers, and operational performance

Database Schema

The project uses a relational database structure consisting of five primary tables: customers, orders, order_items, order_payments, and products. These tables are connected using customer IDs and order IDs, enabling multi-table analytical queries.

Key Business KPIs

Metric	Value
Total Orders	98,666
Unique Customers	95,420
Total Revenue	15.84M
Average Order Value	160.58

Methodology

1. Database Creation – A relational schema was designed using PostgreSQL to store customer, order, product, and payment information.
2. Data Loading – CSV datasets were imported using PostgreSQL's COPY command for efficient bulk loading.
3. Data Validation – Row counts and validation queries were used to confirm successful data

loading.

4. SQL Analysis – Queries were written to analyze revenue performance, customer behavior, product demand, payment patterns, delivery performance, and geographic sales distribution.

Key Insights

- Revenue growth accelerated significantly beginning in early 2017.
- Only 2,997 customers placed repeat orders (~3% repeat purchase rate).
- A small number of product categories generate a large share of revenue.
- Credit cards dominate payment methods.
- Most deliveries occur within estimated delivery windows, with a small percentage delayed.
- Major metropolitan cities generate the highest revenue.
- Freight costs represent a noticeable percentage of product prices.

Business Recommendations

- Improve customer retention through loyalty programs and targeted marketing.
- Optimize logistics operations to reduce late deliveries.
- Focus marketing strategies on top-performing product categories.
- Increase advertising efforts in high-revenue geographic regions.

Tools & Technologies

PostgreSQL, SQL, Relational Database Design, Data Analysis

Future Improvements

- Build an interactive Tableau or Power BI dashboard.
- Perform customer segmentation analysis.
- Implement cohort analysis for customer retention.
- Develop an automated data pipeline for reporting.